

ALDEN NABIL WIBOWO EFFENDY

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SUMMARY

Analytical and growth-oriented Data Science student (GPA: 3.83) skilled in translating raw, disparate datasets into interactive visualizations and automated tools. Leverages programming expertise (Python, R, SQL) to drive Machine Learning and Time Series modeling within economic and renewable energy domains. Proven award-winning project leader blending data-driven decision-making with stakeholder communication to solve complex problems.

EDUCATION

IPB university Bogor, Indonesia

August 2023 – Present

Undergraduate student of Statistics and Data Science

GPA: 3.83

Relevant Coursework: survival analysis, statistical optimization, machine learning technique, introduction to data science, spatial regression, multivariate techniques, time series forecasting methods, data engineering programming, statistical computation, data visualization, regression analysis, probability theory, calculus, relational data management, introduction to management, introduction to economy

HONORS AND AWARDS

3rd Winner and Most Applicable

April – June 2026

Gammafest Data Science Competition 2026

- Engineered an end-to-end predictive machine learning pipeline to forecast international football scores and match outcomes across 121,000+ matches spanning 154 years of historical data (1872–2026)
- Resolved severe feature disparities (47 train vs. 20 test variables) by designing an adaptive chronological Elo rating reconstruction and state-propagation mechanism, successfully extracting 56 predictive metrics with zero data leakage.
- Developed a hybrid modeling framework integrating three parallel LightGBM models (Poisson goal estimators and multi-class outcome classifier) with Optuna Bayesian optimization, reducing the complex non-linear AW-MAE loss from 2.69 to 1.98.

CERTIFICATIONS

- BNSP Data Analyst certified by LSP Tik Global (2026)
- TOEIC test (930/990 score) by ILA IPB (2025)
- Project Management Ready by Project Management Institute (2024)
- Learn Tableau by Working on Exciting Dashboards by Udemy (2025)

PROJECTS

Centralized Normative Database & Benchmarking Platform

June 2026

- Built an end-to-end ETL and standardization pipeline consolidating 17 disparate survey projects (8,800+ observations) into a unified relational database with automated schema validation.
- Developed an algorithmic statistical norm engine to automate historical sensory benchmarking (Mean, Top-Box TB/T2B/T3B, P25/P75 percentiles) across mixed Likert scales.
- Created an [interactive Streamlit analytics platform](#) enabling non-technical stakeholders to perform self-service benchmarking and ingest new survey datasets with automated backups.

Superconductor Critical Temperature Optimization via Elastic Net

January-March 2026

- Engineered and evaluated an Elastic Net regression framework on the UCI Superconductivity dataset (21,197 clean samples, 81 predictors) using Stochastic Gradient Descent (SGD) and custom-built Proximal Gradient Descent (PGD/ISTA) to resolve severe feature multicollinearity.
- Achieved superior predictive accuracy with the tuned SGD model ($R^2 = 0.733$, RMSE = 17.72 K) over PGD ($R^2 = 0.708$) through 5-fold cross-validated grid search, uncovering essential domain descriptors via L_1/L_2 regularization sparsity analysis.

Forecasting Seasonal Commodity Prices with Limited Data: Evaluating ETS vs. LSTM Models

December 2025

- Developed and optimized comparative time-series models (ETS vs. LSTM) over 12 years of historical market data, utilizing Optuna hyperparameter tuning to attain a superior 3.26% MAPE via an optimized ETS framework.
- Designed a reliable early warning system predicting commodity fluctuations 6 months in advance, delivering actionable, data-driven insights for policymakers to optimize inflation control and food security.

ORGANIZATIONAL EXPERIENCES

Society of Renewable Energy, IPB Chapter

January 2025 – January 2026

Head of External and Public Relations

- Secured 10 strategic media partnerships and orchestrated 2 major inter-organizational study events, expanding brand reach to a network of over 100 active community participants.
- Directed external relations strategies and stakeholder engagement, fostering sustainable collaborations that strengthened the organization's network and operational growth.

Gamma Sigma Beta

January 2025 – January 2026

Merchant Specialist – External and Creative Economics Department

- Generated revenue by marketing and liquidating 100+ branded merchandise items through a strategic booth presentation at the Statistics Alumnus Gathering for 200+ attendees.
- Coordinated inter-university benchmarking event with Statistics student associations of Universitas Padjadjaran involving 220 participants.
- Spearheaded the engineering and administration of a central alumni relational database, systematically tracing and archiving graduate records back to the class of 2005

STEM 2024

July 2024 – November 2024

Project Leader

- Upskilled 77 high school students in Design Thinking and AI development, achieving a 4.72/5 program satisfaction score across the entire seminar lifecycle.
- Orchestrated the entire project lifecycle for an introductory Computer Science seminar, leading a 32-person committee and coordinating 3 industry-expert guest speakers from ideation through to final execution.

SKILLS & INTERESTS

Technical Skills: R, Python, SQL, Microsoft Excel, SPSS, Tableau, PowerBI, Figma, Canva,

Soft Skills: Data-Driven Decision Making, Teamwork, Project Leadership, and Event Management, Team Building, Design Thinking

Languages: Indonesia (Native) and English (Fluent)

Interests: Big Data, Machine Learning, Risk Analysis, Time Series Analysis, Sustainability and Renewable Topic, Economics, and Technology,